

Generic Σ_1^2 -maximality

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Abstract

The goal of this note is to present a proof of Woodin's theorem that, under large cardinals, the Continuum Hypothesis is equivalent to the Generic Σ_1^2 -maximality. As a corollary, we get that Σ_1^2 formulas are generically absolute for posets that force CH.

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1 Introduction

We are interested in the following principle of generic maximality.

Definition 1.1. The **Generic Σ_1^2 -maximality** is the statement asserting that for all Σ_1^2 -formulas ϕ , if ϕ can be forced, then it is true. $\square$

The goal of this note is to present a proof, based on [Lar04], of the following theorem.

Main Theorem 1.2 (Woodin). *Suppose there exists a proper class of measurable Woodin cardinals. Then the following are equivalent:*

- a. *Continuum Hypothesis;*
- b. *Generic Σ_1^2 -maximality.*

Proof. The implication $(\Rightarrow)$ follows from Theorem 5.6, while $(\Leftarrow)$ follows from the fact that CH is equivalent to a Σ_1^2 sentence. $\square$

A consequence of this fact is generic absoluteness for Σ_1^2 formulas, conditioned on CH.

Corollary 1.3 (Woodin). *Suppose there exists a proper class of measurable Woodin cardinals and $\mathbb{P}, \mathbb{Q}$ are two posets which both force CH. Then for all Σ_1^2 -formulas ϕ , $V^{\mathbb{P}} \models \phi$ if and only if $V^{\mathbb{Q}} \models \phi$.*

Proof.

- 1° By symmetry, it suffices to show that $V^{\mathbb{P}} \models \phi$ implies $V^{\mathbb{Q}} \models \phi$.
- 2° Let us assume otherwise, i.e. that $V^{\mathbb{Q}} \not\models \phi$. We may assume without loss of generality that $V^{\mathbb{Q}} \models \neg\phi$.
- 3° Let λ be a Woodin cardinal satisfying $\mathbb{P}, \mathbb{Q} \in V_\lambda$, let g be V -generic for $\mathbb{P}$, and let G be $V[g]$ -generic for $\mathbb{Q}_\lambda^{V[g]}$. Since $V[g] \models \phi$, we conclude that $\text{Ult}(V[g], G) \models \phi$.
- 4° The quantifier complexity of ϕ ensure now that $V[g][G] \models \phi$.
- 5° $V \models$ “the powerset of $\mathbb{Q}$ becomes countable after forcing with $\mathbb{P} * \mathbb{Q}_\lambda^{V^{\mathbb{P}}}$ ”. This and Lemma 2.1 imply the existence (in V) of a complete embedding $i : \mathbb{Q} \longrightarrow \mathbb{B}(\mathbb{P} * \mathbb{Q}_\lambda^{V^{\mathbb{P}}})$.
- 6° Let H be the V -generic for $\mathbb{B}(\mathbb{P} * \mathbb{Q}_\lambda^{V^{\mathbb{P}}})$ given by $g * G$, let $h = i^{-1}[H]$, and let H' be a $V[h]$ -generic such that

$$V[g][G] = V[H] = V[h][H'].$$

We have that $V[h][H'] \models \phi$.

- 7° This means that in $V[h]$,

- a. there exists a proper class of measurable Woodin cardinals,
- b. CH holds,
- c. ϕ is Σ_1^2 ,
- d. there exists a poset that forces ϕ .

By Main Theorem 1.2, $V[h] \models \phi$. This contradicts the fact that $V^{\mathbb{Q}} \models \neg\phi$.

□

2 Quotienting Posets

Lemma 2.1. *Suppose*

1. $\mathbb{P}$ is a poset,
2. $\mathbb{B}$ is a complete boolean algebra,
3. it holds in $V^{\mathbb{B}}$ that $\dot{g}$ is a V -generic filter for $\mathbb{P}$.

Then there exists the mapping

$$i : \mathbb{P} \longrightarrow \mathbb{B} : p \mapsto \|\check{p} \in \dot{g}\|_{\mathbb{B}}$$

is a complete embedding. □

Definition 2.2. *Suppose*

1. $\mathbb{B}$ is a complete boolean algebra,
2. $\dot{\mathbb{C}}$ is a complete boolean algebra in $V^{\mathbb{B}}$.

*Then $\mathbb{B} * \dot{\mathbb{C}}$ is the structure*

$$\mathbb{D} = (D, +_D, \cdot_D, -_D, 0_D, 1_D),$$

where

- a. D is the set of all canonical $\mathbb{B}$ -names for elements of $\dot{\mathbb{C}}$ quotiented by the equivalence relation $=_D$ given by

$$\dot{c} =_D \dot{d} : \Longleftrightarrow \|\dot{c} = \dot{d}\|_{\mathbb{B}} = 1,$$

- b. for all canonical $\mathbb{B}$ -names $\dot{c}, \dot{d}, \dot{e}$ for elements of $\dot{\mathbb{C}}$,

$$[\dot{c}] +_D [\dot{d}] = [\dot{e}] \Longleftrightarrow \|\dot{c} +_{\dot{\mathbb{C}}} \dot{d} = \dot{e}\|_{\mathbb{B}} = 1,$$

and similarly for $\cdot_D, -_D, 0_D, 1_D$. □

Proposition 2.3. *Suppose*

1. $\mathbb{B}$ is a complete boolean algebra,
2. $\dot{\mathbb{C}}$ is a complete boolean algebra in $V^{\mathbb{B}}$.

*Then $\mathbb{B} * \dot{\mathbb{C}}$ is a well-defined complete boolean algebra and the mapping*

$$\mathbb{B} \longrightarrow \mathbb{B} * \dot{\mathbb{C}} : b \mapsto \begin{cases} \text{the unique } [\dot{c}], \text{ where } \dot{c} \text{ satisfies} \\ \|\dot{c} = 1_{\dot{\mathbb{C}}}\|_{\mathbb{B}} = b \text{ and } \|\dot{c} = 0_{\dot{\mathbb{C}}}\|_{\mathbb{B}} = -b \end{cases}$$

is a well-defined complete monomorphism.

Proof. See the [Jec03, Lemma 16.3] and its proof. □

Definition 2.4. Suppose

1. $\mathbb{C}$ is a complete boolean algebra,
2. $\mathbb{B}$ is a complete sub-algebra of $\mathbb{C}$.

Then $\mathbb{C} : \mathbb{B}$ is the canonical $\mathbb{B}$ -name for the boolean algebra

$$\mathbb{C} / \{c : \exists b \in \dot{G}_{\mathbb{B}}, c \geq b\}.$$

□

Proposition 2.5. Suppose

1. $\mathbb{C}$ is a complete boolean algebra,
2. $\mathbb{B}$ is a complete sub-algebra of $\mathbb{C}$.

Then $\mathbb{C} : \mathbb{B}$ is a complete boolean algebra in $V^{\mathbb{B}}$ and the mapping

$$\mathbb{C} \longrightarrow \mathbb{B} * (\mathbb{C} : \mathbb{B}) : c \longrightarrow \begin{cases} (\sum \{b \in \mathbb{B} : b \leq c\}, \dot{c}) \text{, where} \\ \dot{c} \text{ is the canonical } \mathbb{B}\text{-name for } [c] \in \mathbb{C} : \mathbb{B} \end{cases}$$

is an isomorphism of boolean algebras.

Proof. See [Jec03, Exercise 16.4].

□

Remark 2.6. The isomorphism of the previous proposition is considered to be canonical.

□

Lemma 2.7. Suppose

1. $\mathbb{C}$ is a complete boolean algebra,
2. τ is a $\mathbb{C}$ -name for a subset of V ,
3. $\mathbb{B}$ is the complete sub-algebra of $\mathbb{C}$ generated by the set

$$\{\|\dot{x} \in \tau\|_{\mathbb{C}} : x \in V\}.$$

Then for all V -generic filters H for $\mathbb{C}$,

- a. $V[\tau^H] = V[H \cap \mathbb{B}]$,
- b. $V[H]$ is a generic extension of $V[\tau^H]$ via the poset $(\mathbb{C} : \mathbb{B})^{H \cap \mathbb{B}}$.

Proof. It is immediate that $V[\tau^H] = V[H \cap \mathbb{B}]$, while the rest follows from Proposition 2.5.

□

Lemma 2.8. Suppose

1. $i : \mathbb{P} \longrightarrow \mathbb{S}$ and $j : \mathbb{P} \longrightarrow \mathbb{Q}$ are complete embeddings of posets,
2. $\mathbb{S}$ forces the existence of a V -generic filter for $\mathbb{Q}$.

Then there exists there exists a complete embedding k such that the diagram

$$\begin{array}{ccc}
 & \mathbb{B}(\mathbb{S}) & \\
 \uparrow & \nwarrow k & \\
 \mathbb{S} & & \\
 \uparrow i & & \\
 \mathbb{P} & \xrightarrow{j} & \mathbb{Q}
 \end{array}$$

commutes, where the unnamed morphism is canonical.

Proof.

1° There exist complete monomorphisms such that the diagrams

$$\begin{array}{ccc}
 \mathbb{S} & \longrightarrow & \mathbb{B}(\mathbb{S}) \\
 \uparrow i & & \uparrow \pi \\
 \mathbb{P} & \longrightarrow & \mathbb{B}(\mathbb{P})
 \end{array}
 \quad
 \begin{array}{ccc}
 \mathbb{Q} & \longrightarrow & \mathbb{B}(\mathbb{Q}) \\
 \uparrow j & & \uparrow \sigma \\
 \mathbb{P} & \longrightarrow & \mathbb{B}(\mathbb{P})
 \end{array}$$

commute.

2° It holds in $V^{\mathbb{B}(\mathbb{P})}$ that $\mathbb{B}(\mathbb{S}) : \pi[\mathbb{B}(\mathbb{P})]$ forces the existence of a V -generic filter for $\mathbb{B}(\mathbb{Q}) : \sigma[\mathbb{B}(\mathbb{P})]$. By Lemma 2.1, there exists a $\mathbb{P}$ -name $\dot{\rho}$ such that it holds in $V^{\mathbb{P}}$ that

$$\dot{\rho} : \mathbb{B}(\mathbb{Q}) : \sigma[\mathbb{B}(\mathbb{P})] \longrightarrow \mathbb{B}(\mathbb{S}) : \pi[\mathbb{B}(\mathbb{P})]$$

is a complete monomorphism.

3° Let k be the complete embedding for which the diagram

$$\begin{array}{ccccc}
 \mathbb{Q} & \longrightarrow & \mathbb{B}(\mathbb{Q}) & \xrightarrow{\simeq} & \sigma[\mathbb{B}(\mathbb{P})] * (\mathbb{B}(\mathbb{Q}) : \sigma[\mathbb{B}(\mathbb{P})]) \\
 \downarrow k & & & & \downarrow (\sigma^{-1})_* \\
 & & & & \mathbb{B}(\mathbb{P}) * (\mathbb{B}(\mathbb{Q}) : \sigma[\mathbb{B}(\mathbb{P})]) \\
 & & & & \downarrow \text{Id}_{\mathbb{B}(\mathbb{P})} * \dot{\rho} \\
 \mathbb{B}(\mathbb{S}) & \xleftarrow{\simeq} & \pi[\mathbb{B}(\mathbb{P})] * (\mathbb{B}(\mathbb{S}) : \pi[\mathbb{B}(\mathbb{P})]) & \xleftarrow{\pi_*} & \mathbb{B}(\mathbb{P}) * (\mathbb{B}(\mathbb{S}) : \pi[\mathbb{B}(\mathbb{P})])
 \end{array}$$

commutes, where

a. the unnamed morphisms are canonical,

- b. $(\sigma^{-1})_* : \sigma[\mathbb{B}(\mathbb{P})] * (\mathbb{B}(\mathbb{Q}) : \sigma[\mathbb{B}(\mathbb{P})]) \longrightarrow \mathbb{B}(\mathbb{P}) * (\mathbb{B}(\mathbb{Q}) : \sigma[\mathbb{B}(\mathbb{P})])$ is defined by asserting that for a canonical $\sigma[\mathbb{B}(\mathbb{P})]$ -name $\dot{c}$ for an element of $\mathbb{B}(\mathbb{Q}) : \sigma[\mathbb{B}(\mathbb{P})]$,

$$(\sigma^{-1})_*([\dot{c}]) := [\sigma_*^{-1}(\dot{c})],$$

where $\sigma_* : V^{\mathbb{B}(\mathbb{P})} \longrightarrow V^{\sigma[\mathbb{B}(\mathbb{P})]}$ is the isomorphism induced by

$$\sigma : \mathbb{B}(\mathbb{P}) \longrightarrow \sigma[\mathbb{B}(\mathbb{P})],$$

- c. $\text{Id}_{\mathbb{B}(\mathbb{P})} * \dot{\rho} : \mathbb{B}(\mathbb{P}) * (\mathbb{B}(\mathbb{Q}) : \sigma[\mathbb{B}(\mathbb{P})]) \longrightarrow \mathbb{B}(\mathbb{P}) * (\mathbb{B}(\mathbb{S}) : \pi[\mathbb{B}(\mathbb{P})])$ is defined by asserting that for a canonical $\mathbb{B}(\mathbb{P})$ -name $\dot{c}$ for an element of $\mathbb{B}(\mathbb{Q}) : \sigma[\mathbb{B}(\mathbb{P})]$, the equivalence class $[\dot{c}]$ is mapped to the equivalence class $[\dot{d}]$, where $\dot{d}$ is the canonical $\mathbb{B}(\mathbb{P})$ -name for $\dot{\rho}(\dot{c})$,
- d. $\pi_* : \mathbb{B}(\mathbb{P}) * (\mathbb{B}(\mathbb{S}) : \pi[\mathbb{B}(\mathbb{P})]) \longrightarrow \pi[\mathbb{B}(\mathbb{P})] * (\mathbb{B}(\mathbb{S}) : \pi[\mathbb{B}(\mathbb{P})])$ is defined by asserting that for a canonical $\mathbb{B}(\mathbb{P})$ -name $\dot{c}$ for an element of $\mathbb{B}(\mathbb{P}) * (\mathbb{B}(\mathbb{S}) : \pi[\mathbb{B}(\mathbb{P})])$, the equivalence class $[\dot{c}]$ is mapped to $[\pi_*(\dot{c})]$, where this last π_* is the isomorphism $V^{\mathbb{B}(\mathbb{P})} \longrightarrow V^{\pi[\mathbb{B}(\mathbb{P})]}$ induced by the isomorphism $\pi : \mathbb{B}(\mathbb{P}) \longrightarrow \pi[\mathbb{B}(\mathbb{P})]$.

4° It is easily verified that k is as required.

□

3 Quotienting Stationary Towers

Definition 3.1. Suppose

1. δ is Woodin,
2. $\theta > \delta$ is regular,
3. $M \prec (H_\theta, \in, \delta)$.

Then M is said to be $\mathbb{Q}_\delta$ -self-generic iff for all $D \in M$ which are predense in $\mathbb{Q}_\delta$, there exists $S \in D \cap M$ such that $M \cap (\cup S) \in S$. $\square$

Definition 3.2. Suppose δ is Woodin. Then $\mathfrak{u}_\delta$ denotes the set of all $M \in [H(\delta^+)]^\omega$ which are $\mathbb{Q}_\delta$ -self-generic. $\square$

Lemma 3.3. Suppose

1. δ is a Woodin cardinal,
2. $\kappa > \delta$ is inaccessible.

Then $\mathfrak{u}_\delta \in \mathbb{Q}_\kappa$ and for all $S \in \mathbb{Q}_\delta$, $S \parallel_{\mathbb{Q}_\kappa} \mathfrak{u}_\delta$.

Proof. This follows from [Kas21, Lemmas 3-XIII]. $\square$

Lemma 3.4. Suppose

1. $\delta < \kappa$ are Woodin cardinals.
2. G is V -generic for $\mathbb{Q}_\kappa$.

Then for all $S \in \mathbb{Q}_\delta^V$, $S \in G$ if and only if

$$\{M \in \mathfrak{u}_\delta^V : M \cap (\cup S) \in S\} \in G.$$

Proof.

1° Let $j := j_G^V : V \longrightarrow \text{Ult}(V, G)$.

2° First, let $S \in G$ and let

$$S^* := \{M \in \mathfrak{u}_\delta^V : M \cap (\cup S) \in S\}.$$

We want to show that $S^* \in G$.

3° This comes down to verifying that $j[H(\delta^+)^V] \in j(S^*)$, i.e. that $j[H(\delta^+)^V] \cap j(\cup S) \in j(S)$ and $j[H(\delta^+)^V]$ is $j(\mathbb{Q}_\delta^V)$ -self-generic in $\text{Ult}(V, G)$.

4° We have that

$$j[H(\delta^+)^V] \cap j(\cup S) = j[H(\delta^+)^V \cap (\cup S)] = j[\cup S] \in j(S),$$

where the last relation follows from the fact that $S \in G$.

5° To see that $j[H(\delta^+)^V]$ is $j(\mathbb{Q}_\delta)$ -self-generic in $\text{Ult}(V, G)$, let $D \in j[H(\delta^+)^V]$ be predense in $j(\mathbb{Q}_\delta^V)$. There exists $D' \in H(\delta^+)^V$ such that $j(D') = D$.

6° By the elementarity of j , D' is predense in $\mathbb{Q}_\delta^V$. Let $T \in D' \cap G$.

7° We have that $j(T) \in D \cap j[H(\delta^+)^V]$ and that

$$j[H(\delta^+)^V] \cap j(\cup T) = j[\cup T] \in j(T),$$

where the last relation follows from the fact that $T \in G$. This shows the self-genericity and verifies the first direction of the lemma.

8° Now, let $S \in \mathbb{Q}_\delta^V$ be such that

$$\{M \in \mathfrak{u}_\delta^V : M \cap (\cup S) \in S\} \in G.$$

We have that

$$S \geq \{M \in \mathfrak{u}_\delta^V : M \cap (\cup S) \in S\},$$

so $S \in G$. This shows the other direction. □

Definition 3.5. Suppose $\delta < \kappa$ are Woodin cardinals. Then the mapping $\mathfrak{q}_{\delta, \kappa}$ is defined to be

$$\mathbb{Q}_\delta \longrightarrow \mathbb{Q}_\kappa \restriction \mathfrak{u}_\delta : S \mapsto \{M \in \mathfrak{u}_\delta : M \cap (\cup S) \in S\}.$$

□

Corollary 3.6. Suppose $\delta < \kappa$ are Woodin cardinals. Then $\mathfrak{q}_{\delta, \kappa}$ is a complete embedding and it holds in $V^{\mathbb{Q}_\kappa}$ that $\mathfrak{q}_{\delta, \kappa}^{-1}[\dot{G}] = \dot{G} \cap \mathbb{Q}_\delta^V$. □

Proposition 3.7. Suppose κ is a measurable Woodin cardinal. Then every real in $V^{\mathbb{Q}_\kappa}$ is $<\kappa$ -generic over V .

Proof.

1° Let x be a canonical $\mathbb{Q}_\kappa$ -name for a real. We will be done if we show in $V^{\mathbb{Q}_\kappa}$ that there exists $\delta < \kappa$ such that

- a. δ is Woodin in V ,
- b. $\mathfrak{u}_\delta^V \in \dot{G}$,
- c. $\dot{x} \in V[\dot{G} \cap \mathbb{Q}_\delta^V]$.

2° This is shown by a density argument. Let $S \in \mathbb{Q}_\kappa$ be arbitrary. We want to find a Woodin cardinal $\delta < \kappa$ and $S' \in \mathbb{Q}_\kappa$ such that $S' \leq S, \mathfrak{u}_\delta$ and

$$S' \Vdash \dot{x} \in V[\dot{G} \cap \mathbb{Q}_\delta^V].$$

3° Since κ is a measurable Woodin cardinal, there exists a Woodin cardinal $\delta \in (\text{rank}(S), \kappa)$ such that

$$V_\delta \prec (V_\kappa, \in, \mathbb{Q}_\kappa, \dot{x}).$$

4° Lemma 3.3 implies that S and $\mathfrak{u}_\delta$ are compatible in $\mathbb{Q}_\kappa$.

5° Let $S' \in \mathbb{Q}_\kappa$ be a common strengthening of S and $\mathfrak{u}_\delta$. It remains to show that

$$S' \Vdash \dot{x} \in V[\dot{G} \cap \mathbb{Q}_\delta^V].$$

6° By the choice of δ , we have that $\dot{y} := \dot{x} \cap V_\delta$ is a canonical $\mathbb{Q}_\delta$ -name for a real.

7° By Corollary 3.6, $\mathfrak{u}_\delta \Vdash \dot{x} = \dot{y} \in V[\dot{G} \cap \mathbb{Q}_\delta^V]$, as required.

□

Proposition 3.8. *Suppose that*

1. N is a transitive model of ZFC,
2. $\delta < \delta^*$ are Woodin cardinals in N ,
3. $\mathcal{P}(\delta^*)^N$ is countable,
4. g is an N -generic filter for $\mathbb{Q}_\delta^N$,
5. x is a set of ordinals,
6. (g, x) is $<\delta^*$ -generic over N .

Then there exists an N -generic filter h for $\mathbb{Q}_{\delta^}^N$ such that $h \cap \mathbb{Q}_\delta^N = g$ and $x \in N[h]$.*

Proof.

1° Let $\mathbb{B}$ be a complete boolean algebra in N of rank $<\delta^*$ and let k be N -generic for $\mathbb{B}$ such that $(g, x) \in N[k]$.

2° Let $\dot{g} \in N^\mathbb{B}$ be such that $\dot{g}^k = g$. Up to restricting $\mathbb{B}$ below a condition, we may assume that

$$N^\mathbb{B} \models “\dot{g} \text{ is an } N\text{-generic filter for } \mathbb{Q}_\delta^N.”$$

3° By Lemma 2.1, the mapping

$$i : \mathbb{Q}_\delta \longrightarrow \mathbb{B} : S \mapsto \|\dot{S} \in \dot{g}\|_\mathbb{B}$$

is a complete embedding in N . It is easily seen that $i^{-1}[k] = g$.

4° Since it holds in $N^{\mathbb{Q}_{\delta^*}^N}$ that $\mathcal{P}(\mathbb{B})^N$ is countable, it follows that

$$N^{\mathbb{Q}_{\delta^*} \restriction \mathfrak{u}_{\delta}}^N \models \text{“there exists an } N\text{-generic filter for } \mathbb{B} \text{.”}$$

5° Lemma 2.8 now implies that there exists in N a complete monomorphism $\sigma : \mathbb{B} \longrightarrow \mathbb{B}(\mathbb{Q}_{\delta^*} \restriction \mathfrak{u}_{\delta})$ such that the diagram

$$\begin{array}{ccc} \mathbb{Q}_{\delta^*} \restriction \mathfrak{u}_{\delta} & \xrightarrow{\pi} & \mathbb{B}(\mathbb{Q}_{\delta^*} \restriction \mathfrak{u}_{\delta}) \\ \uparrow \mathfrak{q}_{\delta, \delta^*} & & \uparrow \sigma \\ \mathbb{Q}_{\delta} & \xrightarrow{i} & \mathbb{B} \end{array}$$

commutes, where the embedding π is canonical.

6° Since $\mathcal{P}(\delta^*)^N$ is countable, there exists an N -generic filter h for $\mathbb{Q}_{\delta^*}^N$ such that $\sigma^{-1}[h] = k$. It is easily seen that h is as required.

□

4 Some Additional Lemmas

Lemma 4.1. *Suppose*

1. δ is a Woodin cardinal,
2. $\theta > \delta$ is regular,
3. $M \prec (H_\theta, \in, \delta)$ is countable,
4. $\alpha \in \delta \cap M$.

Then there exists a countable $N \prec H_\theta$ such that

- a. $M \subseteq N$,
- b. $N \cap V_\alpha = M \cap V_\alpha$,
- c. N is $\mathbb{Q}_\delta$ -self-generic.

Proof. See [Kas21, Lemma 3-XII]. □

Definition 4.2. Suppose

1. κ is an inaccessible cardinal,
2. $\theta > \kappa$ is regular,
3. $M, N \prec (H_\theta, \in, \kappa)$.

Then N is said to **end-extend** M below κ iff $M \subseteq N$ and there exists $\xi < \kappa$ such that

$$M - V_\xi = \emptyset \text{ and } N \cap V_\xi = M \cap V_\xi.$$

□

Lemma 4.3. *Suppose*

1. κ is a measurable cardinal,
2. μ is a normal measure on κ ,
3. $\theta > \kappa$ is regular,
4. $M \prec (H_\theta, \in, \mu)$ is countable,
5. $\gamma = \min \bigcap (\mu \cap M)$.

Then $\text{Hull}(M, \{\gamma\})$ end-extends M below κ .

Proof.

1° Let $x \in \text{Hull}(M, \{\gamma\}) \cap V_{\sup(\kappa \cap M)}$. We want to show that $x \in M$.

2° Let $\alpha \in \kappa \cap M$ be such that $x, \gamma \in V_\alpha$. Then there exists $f \in M$ be such that $f : \kappa \rightarrow V_\alpha$ and $f(\gamma) = x$.

3° Since μ is κ -complete, there exists $y \in V_\alpha$ such that

$$\{\xi < \kappa : f(\xi) = y\} \in \mu.$$

This means that

$$\gamma \in \{\xi < \kappa : f(\xi) = y\},$$

and consequently, $f(\gamma) = y$.

4° Thus, $x = y$ and consequently,

$$\{\xi < \kappa : f(\xi) = x\} \in \mu.$$

5° Since $f, \mu \in M$, it follows that $x \in M$.

□

Lemma 4.4. *Suppose*

1. δ is a Woodin cardinal,
2. G is V -generic for $\mathbb{Q}_\delta$.

Then $\text{Ult}(V, G)$ is closed for $<\delta$ -sequences.

Proof. See [Kas21, Proposition 3-XV].

□

Lemma 4.5. *Suppose*

1. δ is a Woodin cardinal,
2. G is V -generic for $\mathbb{P}_\delta$.

Then $\text{Ult}(V, G)$ is closed for $<\delta$ -sequences.

Proof. This is shown like [Kas21, Proposition 3-XV].

□

Lemma 4.6. *Suppose*

1. $\mathbb{P}$ is a poset,
2. ρ is a regular uncountable cardinal,
3. $\mathbb{P}$ is ρ -c.c.,
4. $\kappa \geq \rho$ is a regular cardinal,
5. g is V -generic for $\mathbb{P}$,
6. C is a club subset of κ in $V[g]$.

Then there exists $\bar{C} \in V$ which is a subset of C and which is a club in κ .

Proof. This is a straight forward generalization of [Jec86, Theorem 1.5].

□

5 Main

Definition 5.1. Suppose

1. N is a transitive model of ZFC^- ,
2. ω_1 is an element of N ,
3. C is a subset of ω_1 .

Then C is an N -slow club iff

- a. every $\alpha \in C$ is Woodin in N ,
- b. C is a club,
- c. for all limit points γ of C , for all $E \in N$ which are club subsets of γ , there exists $\rho < \gamma$ such that $C \cap (\rho, \gamma) \subseteq E$. $\square$

Notation 5.2. For a set X , $\text{tc}(X)$ denotes its transitive collapse. $\square$

Proposition 5.3. Suppose κ is a measurable Woodin cardinal. Then for stationary many $N \prec H(\kappa^+)$,

- a. $\text{tc}(\kappa \cap N) = \omega_1$,
- b. N is $\mathbb{Q}_\kappa$ -self-generic,
- c. there exists a $\text{tc}(N)$ -slow club.

Proof.

1° Let $U \subseteq H(\kappa^+)$. We want to find $N \prec (H(\kappa^+), \in, U)$ satisfying a-c.

2° For a regular $\theta > \kappa$ and a countable $M \prec (H_\theta, \in, \mu)$, we define

$$\gamma_M := \min \bigcap (\mu \cap M).$$

Observe that $(\gamma_M, \kappa) \cap M = \emptyset$.

3° **Claim.** There exists a sequence $(M_\alpha : \alpha < \omega_1)$ such that for all $\alpha < \beta < \omega_1$,

- a. $M_\alpha \prec (H((2^\kappa)^+), \in, U)$ is countable,
- b. M_α is $\mathbb{Q}_\kappa$ -self-generic,
- c. $M_\beta \cap V_{\gamma_{M_\alpha}+1} = \text{Hull}(M_\alpha, \{\gamma_{M_\alpha}\}) \cap V_{\gamma_{M_\alpha}+1}$.

Proof.

1' The sequence is defined by recursion.

2' Let M'_0 be an arbitrary countable elementary submodel of $H((2^\kappa)^+)$ containing U . By Lemma 4.1, there exists a countable $M_0 \prec H((2^\kappa)^+)$ which extends M'_0 and which is $\mathbb{Q}_\delta$ -self-generic.

3' Suppose M_α has been defined and let us define $M_{\alpha+1}$.

4' Let $M'_\alpha := \text{Hull}(M_\alpha, \{\gamma_{M_\alpha}\})$. By Lemma 4.3, M'_α end-extends M_α below κ .

5' By Lemma 4.1, there exists $M_{\alpha+1} \prec H((2^\kappa)^+)$ such that

- a. $M'_\alpha \subseteq M_{\alpha+1}$,
- b. $M_{\alpha+1} \cap V_{\gamma_{M_\alpha}+1} = \text{Hull}(M_\alpha, \{\gamma_{M_\alpha}\}) \cap V_{\gamma_{M_\alpha}+1}$,
- c. $M_{\alpha+1}$ is $\mathbb{Q}_\kappa$ -self-generic.

6' By 4' and 5', for all $\xi < \alpha + 1$,

$$M_{\alpha+1} \cap V_{\gamma_{M_\xi}+1} = \text{Hull}(M_\xi, \{\gamma_{M_\xi}\}) \cap V_{\gamma_{M_\xi}+1}.$$

7' Suppose now that $\alpha < \omega_1$ is a limit ordinal and that $(M_\beta : \beta < \alpha)$ has been defined. We want to define M_α .

8' Let $M_\alpha := \bigcup_{\beta < \alpha} M_\beta$. It is immediate that for all $\beta < \alpha$,

$$M_\alpha \cap V_{\gamma_{M_\beta}+1} = \text{Hull}(M_\beta, \{\gamma_{M_\beta}\}) \cap V_{\gamma_{M_\beta}+1}.$$

9' Let us verify that M_α is $\mathbb{Q}_\delta$ -self-generic. Let $D \in M_\alpha$ be predense in $\mathbb{Q}_\delta$. There exists $\beta < \alpha$ such that $D \in M_\beta$.

10' Since M_β is $\mathbb{Q}_\delta$ -self-generic, there exists $S \in D \cap M_\beta$ such that $M_\beta \cap (\cup S) \in S$.

11' We have the following sequence of relations:

$$M_\alpha \cap (\cup S) = M_\alpha \cap V_{\gamma_{M_\beta}} \cap (\cup S) \tag{1}$$

$$= \text{Hull}(M_\beta, \{\gamma_{M_\beta}\}) \cap V_{\gamma_{M_\beta}} \cap (\cup S) \tag{2}$$

$$= M_\beta \cap V_{\gamma_{M_\beta}} \cap (\cup S) \tag{3}$$

$$= M_\beta \cap (\cup S) \tag{4}$$

$$\in S. \tag{5}$$

□

4° Let $M_{\omega_1} := \bigcup_{\alpha < \omega_1} M_\alpha$ and let $N := M_{\omega_1} \cap H(\kappa^+)$. We want to show that N is as required. Note that it is immediate that $\text{tc}(\kappa \cap N) = \omega_1$.

5° **Claim.** N is $\mathbb{Q}_\kappa$ -self-generic.

Proof.

1' Let $D \in N$ be predense in $\mathbb{Q}_\kappa$. We want to find $S \in D \cap N$ such that $N \cap (\cup S) \in S$.

2' There exists $\alpha < \omega_1$ such that $D \in M_\alpha$. Since M_α is $\mathbb{Q}_\kappa$ -self-generic, there exists $S \in D \cap M_\alpha$ such that $M_\alpha \cap (\cup S) \in S$. In particular, $S \in D \cap N$.

3' We will be done if we show that $N \cap (\cup S) = M_\alpha \cap (\cup S)$.

4' It is clear that $N \cap (\cup S) = M_{\omega_1} \cap (\cup S)$.

5' On the other hand,

$$\begin{aligned} M_\alpha \cap (\cup S) &= M_\alpha \cap (\cup S) \cap V_{\gamma_{M_\alpha}+1} = \\ (M_\alpha \cap V_{\gamma_{M_\alpha}+1}) \cap (\cup S) &= (\text{Hull}(M_\alpha, \{\gamma_{M_\alpha}\}) \cap V_{\gamma_{M_\alpha}+1}) \cap (\cup S) = \\ M_{\omega_1} \cap (\cup S). \end{aligned}$$

6' Thus,

$$N \cap (\cup S) = M_{\omega_1} \cap (\cup S) = M_\alpha \cap (\cup S) \in S.$$

□

6° We now aim to show that the image under the collapse of the set $\{\gamma_{M_\alpha} : \alpha < \omega_1\}$ is such a $\text{tc}(N)$ -slow club. This is verified in the following three claims, which state the “uncollapsed versions” of the properties to be verified: (1) that this set consists of Woodin cardinals of $\text{tc}(N)$, (2) that it is a club, and (3) that this club is N -slow.

7° **Claim.** For all $\alpha < \omega_1$, γ_{M_α} is a Woodin cardinal.

Proof. Since κ is a measurable Woodin, we have that

$$\{\delta < \kappa : \delta \text{ is Woodin}\} \in \mu.$$

Since this set is in M_α , it follows that

$$\gamma_{M_\alpha} \in \{\delta < \kappa : \delta \text{ is Woodin}\}.$$

□

8° **Claim.** For all limit $\alpha < \omega_1$, $(\sup_{\beta < \alpha} \gamma_{M_\beta}, \gamma_{M_\alpha}) \cap M_{\omega_1} = \emptyset$.

Proof.

1' Assume otherwise, i.e. that there exists $\xi \in M_{\omega_1}$ such that

$$\sup_{\beta < \alpha} \gamma_{M_\beta} < \xi < \gamma_{M_\alpha}.$$

2' By the property **c** of Claim 3°, $\xi \in \text{Hull}(M_\alpha, \{\gamma_{M_\alpha}\})$.

3' By Lemma 4.3, $\xi \in M_\alpha$.

4' Hence, there exists $\beta < \alpha$ such that $\xi \in M_\beta$.

5' This means that $\xi < \gamma_{M_\beta}$, which is a contradiction.

□

9° **Claim.** For all limit $\alpha < \omega_1$, for all $E \in M_{\omega_1}$ which are clubs in γ_{M_α} , there exists $\rho < \alpha$ such that

$$\{\gamma_{M_\xi} : \xi \in (\rho, \alpha)\} \subseteq E.$$

Proof.

1' By the property **c** of Claim 3°, $E \in \text{Hull}(M_\alpha, \{\gamma_{M_\alpha}\})$.

2' Hence, there exists $f \in M_\alpha$ such that

- a. $f : \kappa \longrightarrow V_\kappa$,
- b. $f(\gamma_{M_\alpha}) = E$,
- c. for all $\beta < \kappa$, $f(\beta)$ is a club subset of β .

3' Let $\rho < \alpha$ be such that $f \in M_\rho$. We want to show that $\{\gamma_{M_\xi} : \xi \in (\rho, \alpha)\} \subseteq E$.

4' For all $\xi \in (\rho, \alpha)$, we have the following sequence of equivalences:

$$\gamma_{M_\xi} \in E \iff \gamma_{M_\xi} \in f(\gamma_{M_\alpha}) \tag{6}$$

$$\iff \gamma_{M_\alpha} \in \{\beta < \kappa : \gamma_{M_\xi} \in f(\beta)\} \tag{7}$$

$$\iff \{\beta < \kappa : \gamma_{M_\xi} \in f(\beta)\} \in \mu \tag{8}$$

$$\iff \gamma_{M_\xi} \in \{\gamma < \kappa : \{\beta < \kappa : \gamma \in f(\beta)\} \in \mu\} \tag{9}$$

$$\iff \{\gamma < \kappa : \{\beta < \kappa : \gamma \in f(\beta)\} \in \mu\} \in \mu \tag{10}$$

$$\iff \{\gamma < \kappa : \gamma \in j_\mu^V(f)(\kappa)\} \in \mu \tag{11}$$

$$\iff j_\mu^V(f)(\kappa) \in \mu \tag{12}$$

$$\iff \top. \tag{13}$$

The last equivalence holds since by 2', $j_\mu^V(\kappa)$ is a club subset of κ in $\text{Ult}(V, \mu)$.

□

10° Claims 5°, 7°, 8°, and 9° conclude the proof.

□

Proposition 5.4. *Suppose*

1. κ is a measurable Woodin cardinal,
2. $\delta > \kappa$ is Woodin,
3. $\theta \in (\kappa, \delta)$ is inaccessible.

Then there exists $M \prec (H_\theta, \in, \kappa)$ satisfying that

- a. $\text{tc}(\kappa \cap M) = \omega_1$,
- b. there exists a $\text{tc}(M)$ -slow club,
- c. every real is $<\omega_1^V$ -generic over $\text{tc}(M)$.

Proof.

1° We denote by $\mathfrak{U}$ the set of all $M \prec (H_\theta, \in, \kappa)$ such that

- a. $\text{tc}(\kappa \cap M) = \omega_1$,
- b. M is $\mathbb{Q}_\kappa$ -self-generic,
- c. there exists a $\text{tc}(M)$ -slow club.

Proposition 5.3 implies that $\mathfrak{U}$ is stationary in $\mathcal{P}(V_\theta)$.

2° Let G be a V -generic filter for $\mathbb{P}_\delta$ containing $\mathfrak{U}$ and let

$$j := j_G^V : V \longrightarrow \text{Ult}(V, G).$$

By Lemma 4.5, $j[V_\theta] \in \text{Ult}(V, G)$, while it is clear that

$$\text{Ult}(V, G) \models j[V_\theta] \prec (H_{j(\theta)}, \in, j(\kappa)).$$

3° Since j is elementary, we will be done if we show that in $\text{Ult}(V, G)$,

- a. $\text{tc}(j(\kappa) \cap j[V_\theta]) = \omega_1$,
- b. there exists a $\text{tc}(j[V_\theta])$ -slow club,
- c. every real is $<\omega_1$ -generic over $\text{tc}(j[V_\theta])$.

4° Since $j[V_\theta] = [(M : M \in [V_\theta]^\omega)^V]_G$ and $\mathfrak{U} \in G$, Łoś's Theorem implies that

- a. $\text{tc}(j(\kappa) \cap j[V_\theta]) = \omega_1$,
- b. $j[V_\theta]$ is $\mathbb{Q}_{j(\kappa)}$ -self-generic,
- c. there exists a $\text{tc}(j[V_\theta])$ -slow club.

So it remains to show that every real in $\text{Ult}(V, G)$ is $<\omega_1^{\text{Ult}(V, G)}$ -generic over $\text{tc}(j[V_\theta]) = V_\theta$.

5° By 4°a, $\omega_1^{\text{Ult}(V, G)} = \text{tc}(j[\kappa]) = \kappa$.

6° By 4°b, $G \cap \mathbb{Q}_\kappa^V$ is V -generic for $\mathbb{Q}_\kappa^V$. In particular,

$$k : \text{Ult}(V, G \cap \mathbb{Q}_\kappa^V) \longrightarrow \text{Ult}(V, G) : [f]_{G \cap \mathbb{Q}_\kappa^V} \mapsto [f]_G$$

is well-defined and elementary.

7° Since $\omega_1^{\text{Ult}(V, G \cap \mathbb{Q}_\kappa^V)} = \kappa = \omega_1^{\text{Ult}(V, G)}$, it follows that $\text{crit}(k) > \omega_1$. This means that

$$\mathbb{R}^{\text{Ult}(V, G)} \subseteq \mathbb{R}^{\text{Ult}(V, G \cap \mathbb{Q}_\kappa^V)} \subseteq V[G \cap \mathbb{Q}_\kappa^V].$$

8° Proposition 3.7 now ensures that every real in $\text{Ult}(V, G)$ is $<\kappa$ -generic over V . This suffices for the conclusion. □

Proposition 5.5. *Suppose*

1. CH holds,
2. N is a transitive model of ZFC,
3. ω_1 is a Woodin cardinal of N ,
4. every real is $<\omega_1^V$ -generic over N ,
5. there exists an N -slow club.

Then there exists an N -generic filter H for $\mathbb{Q}_{\omega_1^V}^N$ such that $\mathbb{R} \subseteq N[H]$.

Proof.

- 1° Let $x : \omega_1 \longrightarrow \mathbb{R}$ be a surjection, and let C be an N -slow club.
- 2° **Claim.** There exist sequences $\eta = (\eta_\alpha : \alpha < \omega_1)$ and $H = (H_\alpha : \alpha < \omega_1)$ such that for all $\alpha < \beta < \omega_1$,
 - a. $\eta_\alpha \in C$,
 - b. $\eta_\alpha < \eta_\beta$,
 - c. H_α is an N -generic filter for $\mathbb{Q}_{\eta_\alpha}^N$,
 - d. $H_\beta \cap \mathbb{Q}_{\eta_\alpha}^N = H_\alpha$,
 - e. $x_\alpha \in N[H_{\alpha+1}]$.

Proof.

- 1' Let us fix a well-order R of V_{ω_1} . The two sequences are defined by recursion.
- 2' Let η_0 be the least Woodin cardinal of N and let H_0 be the R -least N -generic filter for $\mathbb{Q}_{\eta_0}^N$.
- 3' Let us assume that η_α and H_α are defined and satisfy the recursive hypothesis and let us define $\eta_{\alpha+1}$ and $H_{\alpha+1}$.
- 4' The assumption 4 implies that there exist $\xi \in ((2^{\eta_\alpha})^N, \omega_1)$ and an N -generic filter g for $\text{Col}(\omega, \xi)$ such that $x_\alpha, H_\alpha \in N[g]$. We chose (ξ, g) to be the R -least such pair.

5' Let $\eta_{\alpha+1}$ be the least element of $C \cap (\xi, \omega_1)$. Proposition 3.8 implies that there exists an N -generic filter $H_{\alpha+1}$ for $\mathbb{Q}_{\eta_{\alpha+1}}^N$ such that

$$H_{\alpha+1} \cap \mathbb{Q}_{\eta_\alpha}^N = H_\alpha$$

and $x_\alpha \in N[H_{\alpha+1}]$.

6' For a limit ordinal $\alpha < \omega_1$, we define η_α to be the supremum of the sequence $(\eta_\beta : \beta < \alpha)$ and H_α to be the union of the sequence $(H_\beta : \beta < \alpha)$. It remains to verify the recursive hypothesis.

7' We will be done if we show that H_α is N -generic for $\mathbb{Q}_{\eta_\alpha}^N$. Let $D \in N$ be a predense subset of $\mathbb{Q}_{\eta_\alpha}^N$ and let us show that $H_\alpha \cap N \neq \emptyset$.

8' Note that the set

$$E := \{\xi < \eta_\alpha : N \restriction \xi \prec (N \restriction \omega_1^V, \in, \mathbb{Q}_{\omega_1^V}^N, D)\}$$

is a club subset of η_α that belongs to N .

9' Since C is an N -slow club, E contains a tail of the set $C \cap \eta_\alpha$.

10' In particular, there exists $\beta < \alpha$ such that $\eta_\beta \in E$.

11' This means that

$$N \restriction \eta_\beta \prec (N \restriction \omega_1^V, \in, \mathbb{Q}_{\omega_1^V}^N, D)$$

and consequently, $D \cap \mathbb{Q}_{\eta_\beta}^N$ is predense in $\mathbb{Q}_{\eta_\beta}^N$.

12' We conclude that $H_\beta \cap D \neq \emptyset$, which readily implies that $H_\alpha \cap D \neq \emptyset$. □

3° Let $(\eta_\alpha : \alpha < \omega_1)$ and $(H_\alpha : \alpha < \omega_1)$ witness the claim and let $H := \bigcup_{\alpha < \omega_1} H_\alpha$.

4° **Claim.** H is an N -generic filter for $\mathbb{Q}_{\omega_1^V}^N$.

Proof.

1' It is immediate that H is a filter in $\mathbb{Q}_{\omega_1^V}^N$. Let us verify the N -genericity.

2' Let $D \in N$ be a predense set in $\mathbb{Q}_{\omega_1^V}^N$. Observe that the set

$$E := \{\alpha < \omega_1^V : (N \restriction \alpha, \in, \mathbb{Q}_\alpha, D \cap \mathbb{Q}_\alpha) \prec (N \restriction \omega_1^V, \in, \mathbb{Q}_{\omega_1^V}^N, D)\}^N$$

is a club subset of ω_1 that belongs to N .

3' Thus, there exists $\rho < \omega_1$ such that $C - \rho \subseteq E$.

4' Let $\alpha < \omega_1$ be such that $\eta_\alpha > \rho$. It follows that $\eta_\alpha \in E$.

5' In particular, $D \cap \mathbb{Q}_{\eta_\alpha}^N$ is predense in $\mathbb{Q}_{\eta_\alpha}^N$.

- 6' Since H_α is N -generic in $\mathbb{Q}_{\eta_\alpha}^N$, it follows that $H_\alpha \cap D \cap \mathbb{Q}_{\eta_\alpha}^N \neq \emptyset$.
 7' Consequently, $H \cap D \neq \emptyset$.

□

- 5° Since x is a surjection, the property **e** of Claim 2° ensures that $\mathbb{R} \subseteq N[H]$.
 This concludes the argument.

□

Theorem 5.6. *Suppose*

1. κ is a measurable Woodin cardinal,
2. $\delta > \kappa$ is a Woodin cardinal,
3. CH holds,
4. ϕ is a Σ_1^2 -formula,
5. there exists a poset in V_κ that forces ϕ .

Then ϕ holds.

Proof.

- 1° Let $\mathbb{P} \in V_\kappa$ be a poset that forces ϕ and let $\theta \in (\kappa, \delta)$ be inaccessible. By Proposition 5.4, there exists $M \prec (H_\theta, \in, \kappa, \mathbb{P})$ such that

- a. $\text{tc}(\kappa \cap M) = \omega_1$,
- b. there exists a $\text{tc}(M)$ -slow club,
- c. every real is $<\omega_1^V$ -generic over $\text{tc}(M)$.

- 2° Let

$$\pi : (N, \in, \omega_1, \mathbb{Q}) \longrightarrow (H_\theta, \in, \kappa, \mathbb{P})$$

be the anti-collapse map. Observe that there exists an N -generic filter h for $\mathbb{Q}$.

- 3° Let us fix one such filter h . We have that $N[h] \models \phi$.

- 4° **Claim.** Every real in $<\omega_1$ -generic over $N[h]$.

Proof. Let x be an arbitrary real. Then there exists a real y that codes (x, h) . By the choice of M , y is $<\omega_1$ -generic over N . Since $h \in N[y]$, Lemma 2.7 implies that $N[y]$ is a generic extension of $N[h]$ via a poset in $N \upharpoonright \omega_1^V$. The conclusion now follows from the fact that $x \in N[y]$. □

- 5° **Claim.** There exists an $N[h]$ -slow club.

Proof.

- 1' Let C be an N -slow club.
- 2' Since $\mathbb{Q} \in N \restriction \omega_1^V$, there exists $\rho < \omega_1$ such that ρ is a regular uncountable cardinal and $\mathbb{Q}$ has ρ -c.c. in N .
- 3' Let $C' := C - \rho$. We want to show that C' is a $N[h]$ -slow club.
- 4' The non-trivial thing to verify is that for a given limit point η of C' and a club subset $E \in N[h]$ of η , there exists $\alpha < \eta$ such that $C' \cap (\alpha, \eta) \subseteq E$.
- 5' Since $\eta \in C$, it is regular in N .
- 6' Since in N , $\mathbb{P}$ is ρ -c.c. and $\eta \geq \rho$ is regular, Lemma 4.6 implies the existence of $\bar{E} \in N$ which is club in η and contained in E .
- 7' Since C is an N -slow club, there exists $\alpha < \eta$ such that $C \cap (\alpha, \eta) \subseteq \bar{E}$.
- 8' It is now clear that $C' \cap (\alpha, \eta) \subseteq E$.

□

- 6° Proposition 5.5, together with Claims 4° and 5°, implies the existence of an $N[h]$ -generic filter H for $\mathbb{Q}_{\omega_1^V}^{N[h]}$ satisfying that $\mathbb{R} \subseteq N[h][H]$.
- 7° Lemma 4.4 ensures that $\mathbb{R} \subseteq \text{Ult}(N[h], H)$.
- 8° Since $N[h] \models \phi$, we conclude that $\text{Ult}(N[h], H) \models \phi$. Since $\mathbb{R} \subseteq \text{Ult}(N[h], H)$ and since ϕ is Σ_1^2 , it follows that ϕ holds in V .

□

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